

## **CLASSWIDE LIST – CONCEPTS, METADATA, CONSTRAINTS, FAB 2026**

### **CONCEPTS**

Abstract patterns,  
Abstraction  
Actions,  
Adaptation  
Administrative decision-making,  
Adversarial search,  
Agent specialization,  
Agent-environment distinction,  
Agentic environment  
Aggregate rewards,  
Algorithm  
Alleles,  
Ambiguous feedback  
Arc consistency,  
Artificial selection  
Aspiration Level (satisfactory threshold).  
Autoencoders  
Backtracking  
Backward propagation (backprop),  
Bandwidth,  
Bayesian networks / Bayesian inference  
Beliefs (probabilities)  
Binary constraint,  
Binary variables (Boltzmann),  
bit  
Bit strings,  
Boltzmann Machines (Deep Learning)  
Branches,  
Breadth-first search,  
Brittleness  
Brute force  
Building Blocks  
Causal nets / Bayesian networks  
Cells/Objects (symbolic containers)  
Channel capacity,  
Child variable,  
Chromosome (bit string representation)  
Coding libraries,  
Cognitive representations,  
Coherence  
Combinatorial complexity  
Combinatorial Explosion  
Commonsense knowledge,  
Communication systems,  
Complete assignment  
Complex problems,

Compression,  
Concept Map  
Conceptual Units / Categories  
Conditional probabilities  
Configuration of agents,  
Connections,  
Consistent assignment,  
Constraint network  
Constraint propagation,  
Constraint Satisfaction Problems (CSPs)  
Constraints (Unary, Binary, Higher-order)  
Contribution function  
Convenient fictions  
Cost function,  
Costs (soft constraint violation),  
Credit Assignment  
Crossover  
Cumulative expected value,  
Curse of dimensionality  
Data types (programming)  
Database integrity  
Decision models  
Decoding,  
Decomposition,  
Delayed feedback  
Delegation,  
Deliberate practice  
Depth-first search,  
Differences (current vs goal),  
Digital abstraction,  
Directed acyclic graph (DAG),  
Division of labor  
Domain  
Effective procedure  
Effectuation  
Encoding  
Entropy minimization (Boltzmann),  
Entropy,  
Environment  
Evaluation function  
Events  
Evolutionary operators,  
Evolutionary Strategies  
Expert Systems  
Explicit knowledge,  
Exploitation  
Exploration  
Exploration-exploitation tradeoff,  
External representations

Feedback,  
Find the best solution  
Find-all-solutions goal,  
Find-any-solution goal,  
Find-best-solution goal,  
Fitness  
Fitness function,  
Fitness measure  
Forward-checking  
Gains to specialization  
Game tree  
Gene/Allele  
General Problem of Intelligence  
General Problem Solver (GPS),  
Generate-and-test  
Genes,  
Genetic algorithms (GAs)  
Genetic sequences,  
Genotype,  
Giant Strides (logical evidence-based moves)  
Global optimum  
Goal State  
Goal-directed agents  
Goal-directed agents,  
Goals,  
Gradient Descent ,  
Graph structure,  
Hard constraints,  
Heuristics  
Hierarchical management,  
Hierarchy  
Higher-order constraint,  
Hill-climbing (incremental trial-and-error)  
IID (independently and identically distributed),  
IID vs. OOD Problems  
Implicit parallelism  
Infinite Regress  
Information  
Information Processing  
Information theory,  
Informed search  
Initial state  
Intelligence  
Interdependence of actions,  
Intermediate feedback  
Intermediate State  
Internal representations,  
Knowledge Acquisition Bottleneck  
Knowledge base

Knowledge Representation and Reasoning (KRR)  
Learning in a Society / The Management Hypothesis  
Learning rate  
Learning  
Lists,  
Local peaks (local maxima),  
Local vs. Global Optima  
Long-jumping (moving across peaks)  
Management hypothesis,  
Market mechanism,  
Maximize rewards  
Meaningful abstraction,  
Means-Ends Analysis (General Problem Solver)  
Mechanization of thought,  
Memory management,  
Memory of action sequences,  
Mendelian approach  
Mental models (decision models)  
Mesa landscape  
Mesa topology  
Message-passing system,  
Minimal effective procedure,  
Modifications,  
Modularity  
Modules  
Mutation  
Myopia (seeing only locally)  
N-ary constraint,  
Natural language prompting,  
Natural selection,  
Node  
Nodes (constraint),  
Noise  
Novel configurations  
Objects (propagator),  
OOD (out-of-distribution),  
Operators  
Optimization goal,  
Organizational channel (hierarchy),  
Organizational learning,  
Organizational memory,  
Organizational programs  
Paradigms  
Parent variable  
Partial assignment,  
Path Dependence (early actions affecting the end result)  
Paths (action sequences).  
Pattern matching  
Pattern recognition,

Performance Landscape  
Physical Symbol System Hypothesis  
Physics hypothesis,  
Planning (lookahead),  
Plans / Programs  
Policy  
Population of chromosomes,  
Population-level learning,  
Possible worlds,  
Preferences (soft),  
Principle of locality  
Probabilities,  
Probability distribution (learned),  
Procedural Complexity  
Processing capacity,  
Processing complexity  
Processing power,  
Productions (if-then rules),  
Programs,  
Propagation (sending signals to related objects).  
Propagator Model  
Pruning  
Random sampling,  
Random variables  
Reinforcement learning  
Relational databases  
Relations (possible worlds)  
Representation / Schemas  
Representation learning  
Retention  
Reward hypothesis  
Rewards,  
Rugged landscape  
Satisficing  
Satisfy a set of constraints  
Schema  
Schema (genetic),  
Scope  
Search  
Search space,  
State space,  
Search Tree  
Selection,  
Semantic information  
Shannon entropy  
Shannon Information (Syntactic)  
Shannon information,  
Shared language

Short-term / Long-term Memory Capacity  
Signal propagation (cell analogy),  
Signal-to-noise ratio  
Signals,  
Simon Information (Meaning, Symbols)  
Simon Search (open-ended means-ends analysis).  
Single-peaked landscape,  
Smoothing topology,  
Society of memories  
Society of mind hypothesis  
Soft constraints,  
Solution,  
Sparse feedback  
Sparse Learning  
Spatial Interdependence  
State  
State Space  
State transitions,  
State,  
Statistical distribution,  
Statistical redundancy removal,  
Subgoals,  
Superstitious Learning  
Superstitious learning  
Symbol Manipulation  
Syntactic information  
Tacit knowledge,  
Task environment  
Temporal differencing (TD),  
The Digital Abstraction  
Thinking machine,  
Trial-and-error feedback  
Trial-and-error,  
Tuples,  
Turing machine  
Unary constraint,  
Uninformed search,  
Unitary actor assumption,  
Utility Function (numerical reward)  
Value function,  
Variables  
Variation,  
Von Neumann Search (adversarial games)  
VSR  
Watchmaker parable  
Weights (Boltzmann = correlations),

## **RELATIONS / METADATA**

- Section number

- "Comprehension", "prerequisite" metadata: monitoring the state of understanding — this is similar to the database of remaining possible values in Sudoku.

- **People / Institutions:** (Arthur Samuel, IBM, Marvin Minsky, MIT AI Lab, Richard Sutton, Andrew Barto, University of Massachusetts Amherst, Richard Bellman, RAND Corporation, Herbert Simon, Carnegie-Mellon, Norbert Wiener, Holland (1975), Levitt & March (1988), Argote & Epple (1990), Thomke (2020), Cyert & March (1963), Denrell Fang & Levinthal (2004), Lave & March (1975), Claude Shannon (1916 2001), Herbert Simon (1916 2001), Bell—  
Labs, AT&T, Allen Newell, Saleema Amershi, Microsoft Research, Richard Sutton, Ludwig von Boltzmann, Cantwell-Smith (2021), Adam Smith, Charles Babbage, Alan Turing (1936), John von Neumann, Herbert Simon, Allen Newell, Demis Hassabis, Google DeepMind, Arthur Samuel, Prietula and Simon (1989), Charles Darwin, John Holland (University of Michigan), Gregor Mendel, Gerald Sussman (MIT), Marvin Minsky (in ES context), Herbert Simon (watchmaker), Babbage (2nd watchmaker analogy); Judea Pearl, Geoff Hinton (Nobel Prize), Feigenbaum (1964), Hubert Dreyfus (1965), Wittgenstein, Prietula and Simon (1989), Ludwig von Boltzmann, Newton (classical mechanics reference), Minsky (1961) in representation context, Russell & Norvig (toothache example))

- **Examples / Metaphors:** (Shipbuilding company, App development team, Budgeting committee, Pharmaceutical company, A/B testing, Business units, Quarterly budgets, KPIs (key performance indicators), Customer journey, Milestones, T-maze, Mouse, Cheese, Left/right actions, Goal-box, Weighted coin (70/30), CEO agent, Builder agent, Add agent, Grasp agent, Move agent, Muscle fibers, Wooden blocks T-maze, Hedge maze, Containment property, Geoffrey (builder human), Net present value (NPV), Cash flows, Time value of money, Checkers game, Number of pieces, Distribution of pieces, Checkers proxies, Theseus robot (1950), Maze navigation, Electrical circuits, Telephone network routing, Pilot cockpit example, Altimeter reading, Air traffic control, Fiber optic cables, 5G network, TV channel bandwidth, Coin toss (fair coin),

Weighted coin edge case, Artificial brain, Neuron synapses (100 trillion), Watchmaker parable (1000 parts), 100 modules of 10 parts, Interruption probability 1%, Food ordering model, French language model, Apple VisiCalc, Microsoft Excel (financial projections), English word distribution, Laptop design (screen, CPU, memory, keyboard), Contribution of CPU to memory compatibility, Plant breeding (Mendel), Pin factories competition (Babbage/Darwin), Hippos in tropics, Penguins in Antarctica, Biological cells, Watch components (hour hand, spring), Mechanical watch assembly, Factory analogy (propagator), Market forces selection, Firm strategy design, Organization design, Product design as genotype; Sudoku (9x9 grid, 81 squares, numbers 1-9), Row constraint, Column constraint, 3x3 grid constraint, "All different" constraint, Safe combination comparison (81! vs 6 numbers), Crossword puzzle (less/richer propagation), Car exterior/interior color (constraint example), Toothache-cavity diagnosis, Dental history variable, Tooth-brushing habits variable, Pin factory scheduling (CSP), Task 1 Make pin base, Task 2 Make pinhead, Task 3 Assemble pin, Temporal order constraint (Task 1+2 before Task 3), Parallel task execution (2 workers 2 mins), 1000-part product → scheduling, KAIST to Incheon Airport map (train route pruning), Green train lines (don't take), Algebraic notation (17th century), Double-entry bookkeeping (15<sup>th</sup> century), Engineering drawings (General Motors 1950s, Tamron 1970s), VisiCalc (Apple 1979), Microsoft Excel financial projections,

Relational databases / SQL, databases, CAD software (2D/3D), Filing cabinet vs relational database, Mycin (expert system, bacterial infections, ~200 rules), Image classification (Boltzmann output), LLM text generation (Boltzmann output);

- Each concept should be related to the course notes name
- Concepts can be defined in terms of each other, etc.
- To understand a subsection, the student should first understand its parent section.
  - o (For example, a student's understanding of Simon Search is strictly constrained by prior definitions of "States" and Heuristics and Plans". Understanding of "Basic Structure of Search Strategy " and "Von Neuman Search and Simon Search" naturally results in a conclusion of "Understanding of using search process to make information processing more efficient".)
- If a section is selected, its keywords become active and available for propagation.
- If a keyword is selected, the student can propagate to the section(s) where it is explained.
- The goal state is complete when all important sections are covered by relevant keywords.
- if (understand contents of section n, where  $n < \text{Value}$ ) then (can understand contents of Value)
  - o if (understand definition of keyword k) then (can understand contents of section s,
  - o where k appears in s)
  - o if (understand description of keyword k) then (can understand contents of section s,
  - o where k appears in s)
- Definitions; Mechanisms; Examples; Comparison tables; Mapping (org. analogy); Assessment unit (quiz etc.).
- Question type: fill in the blank / multiple choice
- Concept difficulty status (like remaining possible values)
- Sections
- Concept abstraction level
- Depth of description (short, medium, detailed, i.e., in word length).
- Strongly related concepts should include a *reference link* to the other.
- Level of abstraction
- Whether the concept has a usable definition in the concept map
- Defined / Partially defined / Undefined
- IS-A | PART-OF | CAUSES | CONTRASTS-WITH | EXAMPLE-OF
- Node Clarity Whether the node's explanation is self-contained without the source
- Tested / Not tested

## CONSTRAINTS

**Compare to text or slides → very high propagation; but low elaboration tolerance!**

- Each concept can optionally have (support/counter) arguments.
- If two arguments support a concept, these two arguments cannot contradict one another
- Pairwise weights on concepts OR sentences.
- Each Concept must be linked to at least one Definition.
- Each Definition must belong to one Concept.
- Each Example must be linked to at least one or more Concept.
- Each Quiz item must be linked to at least one or more Concept.
- Parent-child
- Paradigm distinction; constraints on definitions and mechanisms.
- Hierarchy: a more abstract concept covers > 2 concepts. Hierarchical Constraints: Relation between the higher (more abstract) concept and the lower (more specific) concept (Which indicates inclusion relationship).



- A is more abstract than B
- A requires B as a foundation
- A enables or extends to B
- A contrasts with B (bidirectional)
- A extends into B
- A leads to the discovery of B
- A is a mechanism/instance of B
- Causal
- Definitional: combines other concepts.
- Attributes
- Similarity: association, aggregation, composition, dependencies
- Keywords must be linked to one or more concepts
- Phrases: Directly linked to one of the key concepts
- Concepts  $\leftarrow$  keyword
- Concept-concept
- Abstraction Ordering: If concept A IS-A or PART-OF concept B, then AbstractionLevel(A) must be strictly more concrete than Abstraction Level(B). Example: 'hill climbing' (Concrete) IS-A 'heuristic search' (Intermediate) IS-A 'search' (Abstract).
- Coverage Before Testing: If QuizRelevance(X) = Tested, then Coverage(X) = Covered. A concept cannot appear on the quiz if it is missing from the network.
- Clarity Before Linking: A node X can be the source of a relation edge only if NodeClarity(X) = Clear. An unclear node must be refined before it can constrain others.
- Module–Abstraction Consistency: Concepts in Module A or B (reading / compression) must have AbstractionLevel = Concrete or Intermediate. Abstract concepts belong to Module C or D.
- Symmetry of CONTRASTS-WITH: Relation(A, B) = CONTRASTS-WITH  $\blacksquare$  Relation(B, A) = CONTRASTS-WITH. Contrast is always bidirectional.
- Prerequisite/Causal Constraints : Certain mechanisms require foundational knowledge to process.
- Comparative Constraints: When parallel concepts (like the three types of networks) are learned, they share structural constraints.
- Depends on
- Contrasts with
- Comparison tables should show both keywords and examples
- Text units must map to at least one concept
- Hierarchical: abstract  $\rightarrow$  concrete
- Causal: if x then y
- Comparative (A vs. B): each concept must connect
- 1) Each concept should be one idea only
- 2) It should have a clear meaning
- 3) It should not be confusing or ambiguous
- 4) It should be a standard term we actually use in the course
- 5) Concepts in a list should be at the same level (not too mixed)
- 6) We should understand it on its own 7) 8) 9) No repeating the same concept twice
- It should connect to other concepts (cause, part of, etc.)
- We should be able to break it into smaller parts if needed
- 10) It should relate to something real (formula, process, or measurable thing)
- **Nodes in same subtree:** The nodes included in the subtree must provide the user with sufficient understanding. If it is determined that the user has not achieved sufficient understanding after

solving a quiz on the content of the subtree, the subtree must be modified to include more nodes and specific details.

- Variables representing a student's proficiency, reading strategy, and note type are linked by propagation rules that automatically restrict or unlock options based on quiz performance and self-testing outcomes. Prerequisite relationships between concepts are enforced such that unassessed foundational concepts block dependent ones, and any unresolved concept blocks module output entirely.